

ELIAS HAKENSO

Data Analyst | AI Engineering

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SUMMARY

Data analyst and AI engineer with 3.5 years validating and reporting on a 25,000+ record institutional dataset in Python, Excel, and SAS. Built a **Power BI** gap analysis whose finding went to King County's Language Access Program, and a real-time voice QA harness that found **8 defects** in a live healthcare AI agent. Writes the rules before touching the data.

EDUCATION

M.S. in Data Science (Executive Format, Online), University of the Cumberlands Expected May 2028

B.S. in Computer Information Technology, Minor in Mathematics Aug 2025

Minnesota State University, Mankato, MN | GPA 3.7, Magna Cum Laude, Alpha Lambda Honor Society

Applications of AI Engineering Certificate, CodePath.org Aug 2026

- Six AI projects (RAG, agents, fine-tuning, guardrails), a Docker codebase contribution; open source capstone in progress.

SELECTED PROJECTS

Language Access in South King County: Service Gap Analysis | Python, GeoPandas, Census API, Power BI, DAX

- Measured home-language demand (**36,775 families, 140 languages**) against public-service supply, scoring **6,272** agency-by-language pairs across **51 agencies in 7 sectors**, each backed by an evidence link.
- Found that King County's legally binding translation list, built on 2016 data, omits the region's second most common home language (Dari, **3,040 families**); sent the finding as a memo to the county.
- Rejected a standard Census apportionment method after validation showed **7.1 percentage points** mean absolute error; published a 4-page **Power BI** report on an **8-table star schema with 23 DAX measures**.

Patient-Sim: Voice Agent QA Harness | Python, Twilio, OpenAI Realtime API, WebSockets

- Built a voice bot that phones a live healthcare AI agent as a realistic patient, bridging **Twilio Media Streams to the OpenAI Realtime API** over WebSockets with G.711 μ -law end to end (zero transcoding).
- Ran **10 YAML scenarios** as pure data (no code changes per test) across **11 recorded, transcribed calls**; documented **8 defects**, with guards for a live system (number assertion, 330-second cutoff, serial calls).

Provenance Guard: AI Content Detection Service | Flask, Groq (Llama 3.3 70B), Python

- Fused a Llama 3.3 70B classifier and a stylometric heuristic (weighted **0.65 to 0.35**) behind four Flask endpoints, with a **0.40 to 0.65** uncertain band and a disagreement penalty.
- Documented a false-positive failure mode on non-native English writing; shipped appeals, audit logs, and rate limits.

TakeMeter: Fine-Tuned Classifier and Evaluation | Hugging Face, DistilBERT, Groq

- Labeled a 200-example dataset under written rules and benchmarked fine-tuned DistilBERT against a zero-shot LLM baseline (precision, recall, F1, confusion matrices); reported that the baseline won (76.7% vs 70%).

EXPERIENCE

Data Entry Specialist, Minnesota State University, Mankato, MN Dec 2021 to May 2025

- Maintained and validated a 25,000+ record student dataset with repeatable checks that caught discrepancies early.
- Automated recurring cleaning and reporting in Python and Excel (formulas, pivot tables); ran statistical analysis in SAS and explained findings to non-technical staff.

Co-Founder, BrightPath Flexible Tutoring, Greater Seattle, WA May 2025 to Present

- Run diagnostic assessments, individual learning plans, and monthly progress reports for K-12 students.

SKILLS

Data and BI: SQL, Python (pandas, NumPy, GeoPandas, scikit-learn), Power BI (DAX, star schema), Excel (pivot tables), SAS

Applied AI: LLM APIs (OpenAI Realtime, Groq), prompting, agents, RAG (ChromaDB), fine-tuning, guardrails, LLM-as-judge

Evaluation and Methods: precision, recall, F1, baseline benchmarking, data validation, statistical and sensitivity analysis

Engineering: Flask, REST APIs, WebSockets, Twilio, Docker, Git, pytest; bilingual in English and Amharic